# AI BASED ASSISTANCE FOR HOME AUTOMATION


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***Abstract*—This project presents a dual-mode home automation system integrating voice commands and a web dashboard for seamless control of household appliances using ESP32 and Raspberry Pi. Designed with a focus on accessibility, offline operation, and affordability, the system empowers users—especially the elderly and differently-abled—to operate devices through local speech recognition (using Vosk and Python) and a browser-based interface hosted on the ESP32. It supports real-time control of fans, lights, and email messaging without reliance on cloud platforms or smartphones. The ESP32 manages device switching, web hosting, and email transmission, while the Raspberry Pi processes voice input and provides audio feedback. Field testing confirmed over 85% speech accuracy, sub-second response times, and stable performance in network-variable environments. The system's modular and scalable architecture makes it ideal for both practical deployment and educational purposes, highlighting a significant improvement over prior models which often lacked offline functionality, user feedback, and dual control modes.**


## I. Introduction

In the rapidly evolving landscape of smart automation, the need for intuitive, hands-free control of household devices has grown significantly. Traditional systems often depend on mobile applications and proprietary hardware, limiting accessibility for users with mobility or technical constraints. In this context, integrating voice recognition with web-based interfaces over IoT frameworks introduces a compelling solution for modern smart homes. This project introduces a dual-mode control system utilizing the ESP32 microcontroller, offering both voice command input and browser-based device management.

The core system allows users to control appliances like fans and lights and send custom email messages without the need for a smartphone or PC. Voice commands are captured through a Raspberry Pi or PC, processed using speech recognition libraries, and transmitted to the ESP32 via Wi-Fi. Simultaneously, a web dashboard hosted on the ESP32 provides manual control with real-time responsiveness. These features make the system versatile, cost-effective, and highly usable in homes, offices, or remote areas where manual operation may be inconvenient or unsafe.

The proposed solution is especially useful in enhancing the quality of life for the elderly and physically challenged, by providing a simple and contactless means of communication and control. It also serves as an educational platform for learning about embedded systems, wireless communication, and speech interfaces. Through this integration, the project not only addresses technological gaps but also emphasizes inclusivity, adaptability, and practicality in smart home automation.

## II. Literature Survey

The system uses Natural Language Processing (NLP) and IoT to process user commands through Wi-Fi. It interprets natural language voice inputs for device control. However, it may not function accurately in noisy environments or with varied accents [1].

The automation framework integrates AI and IoT for smart control and high-level home security. It features real-time monitoring using Wi-Fi and cloud connectivity. Key concerns include data security and potential vulnerabilities in IoT-based systems [2].

An AI-powered model enables home automation using voice recognition and fingerprint-based dual-level authentication. Real-time control ensures secure and responsive operation. Limitations include sensitivity to environmental noise, dependency on power/network, and limited scalability [3].

The design supports voice-based automation using Google Assistant and real-time device control via mobile apps. It is compatible with commonly available hardware and services like Blynk. However, it lacks offline capability and raises concerns about cloud privacy and constant internet reliance [4].

The model integrates NLP and machine learning to recommend intelligent device actions. It supports real-time monitoring and voice interaction for smart environments. Limitations include inaccurate speech-to-text conversion, security concerns, and NLP processing issues [5].

The system connects IoT devices with Google Assistant for voice-based smart home control. Real-time feedback and monitoring improve interactivity. However, it struggles with recognition accuracy in noisy settings and limited range of voice input [6].

Home automation is achieved using ESP32 microcontroller, cloud-based monitoring with Firebase, and sensor integration. The system offers voice-command capabilities and remote interaction. Issues include basic security, dependence on internet connectivity, and limited third-party integration [7].

A timer switch is automated using voice input processed through an AI assistant. Integration with IFTTT and Blynk enables customized scheduling via speech. Drawbacks include limited functionality duration, restricted system compatibility, and internet reliance [8].

Voice control is combined with mobility assistance, enabling a smart wheelchair system that also automates home devices. Custom wake words enhance accessibility and hands-free control. Challenges include poor performance in noisy environments, limited scalability, and reliance on stable internet [9].

Speech commands are interpreted for real-time home automation using a cloud-connected platform. The system integrates hardware and software for remote device management. Limitations include poor recognition quality, internet dependency, and absence of a display interface [10].

## III. Summary of the Literature Survey

The literature survey reveals that recent advancements in smart home automation heavily focus on the integration of artificial intelligence, voice recognition, and IoT technologies. Most systems utilize microcontrollers like ESP32 or Raspberry Pi and depend on cloud platforms such as Google Assistant, Firebase, or Blynk for real-time control. While these approaches improve user convenience, they also exhibit common limitations, including dependency on stable internet connectivity, poor performance in noisy environments, lack of offline functionality, and minimal support for communication features like email alerts. Many systems fail to include fallback mechanisms or dual-mode control options, which are essential for reliability and inclusivity—especially for elderly or non-technical users. Innovations like gesture-based control, emotion detection, and biometric authentication have been explored but are often limited by hardware complexity, scalability issues, or high costs. The proposed project addresses these gaps by introducing a dual-mode system with offline voice recognition, local web dashboard control, and integrated email functionality, thereby offering a more practical, affordable, and inclusive solution for smart home automation.

## IV. Scope of the Work

The scope of this work is to design and implement a dual-mode smart automation system that provides both voice-based and web-based control over household appliances using affordable and accessible hardware such as the ESP32 and Raspberry Pi. The system is intended to operate efficiently in both online and offline environments, making it suitable for rural and low-connectivity areas. It supports essential automation features such as device switching, real-time status monitoring, and email communication, all without the need for smartphones, cloud services, or high-end computing infrastructure. The project aims to benefit a wide range of users, including the elderly and physically challenged, by offering an intuitive and contactless interface. Furthermore, it serves as a scalable educational platform for exploring IoT, embedded systems, and AI-based interaction technologies, thereby enabling future expansion through integration with additional sensors, mobile applications, and security mechanisms.

## V. Objective

1.To develop a dual-mode smart home automation system that supports both voice command and web-based control.

2.To implement offline speech recognition using Python libraries like Vosk on a Raspberry Pi.

3.To allow the ESP32 microcontroller to send custom emails via SMTP without the need for a smartphone or PC.

4.To design a user-friendly web dashboard hosted locally on the ESP32 for real-time device control and status updates.

5.To ensure the system is accessible to elderly and non-technical users through intuitive voice interaction and feedback mechanisms.

6.To avoid dependence on cloud services, enabling reliable operation even in low or no internet environments.

## VI. Methodology

The proposed system follows a modular and layered architecture to enable dual-mode control—via voice commands and a web dashboard. Voice input is captured through a microphone connected to a Raspberry Pi, where offline speech recognition is performed using Python libraries like Vosk. The interpreted command is transmitted over Wi-Fi to the ESP32 microcontroller, which processes it to switch connected appliances like lights or fans through relay modules. Simultaneously, the ESP32 hosts a browser-accessible dashboard using the ESPAsyncWebServer library, enabling manual control and real-time status updates. For communication, the ESP32 is equipped with SMTP support to send custom emails without the need for a PC or smartphone. Additionally, audio feedback is provided through a speaker using text-to-speech functionality on the Raspberry Pi, enhancing user interaction. The system is built using low-cost, widely available components and optimized for offline operability, making it ideal for smart environments in rural areas or for users with limited technical expertise.

*Block Diagram Description

User Voice The system begins with the user speaking a command (e.g., ”Turn on light”).

Microphone A microphone captures the user’s voice and sends the audio signal for processing.

1.ESP32 NodeMCU and Raspberry Pi 3 This is the central processing unit for the system (in your case, primarily the Raspberry Pi 3). It:

2. Converts speech to text using offline speech recognition (e.g., Vosk).

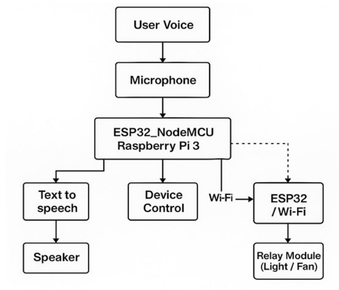


Fig. 1: Block diagram

3. Determines the intended command (e.g., control a device or send feedback).

4. Communicates with the ESP32 for further execution.

5. Text-to-Speech $\rightarrow$ Speaker If audio feedback is needed, the recognized command is converted to speech using a text-to-speech engine. The speaker then announces a confirmation, such as "Fan turned on".

6.Device Control If the voice command involves controlling a device (like a fan or light), the Raspberry Pi sends the command to the ESP32 via Wi-Fi.

## VII. SOFTWARE REQUIREMENT

he proposed system utilizes a combination of open-source and lightweight software tools to ensure affordability, ease of development, and offline functionality. The Arduino IDE is used to program the ESP32 microcontroller, allowing configuration of GPIO pins, Wi-Fi connectivity, and control logic for relay modules. For voice processing and speech recognition, the Raspberry Pi runs Python 3.x scripts utilizing libraries such as Vosk and SpeechRecognition to perform offline voice-to-text conversion. Text-to-speech feedback is implemented using Python's pyttsx3 or similar libraries. The web interface is developed using HTML and CSS, and is hosted locally using the ESPAsyncWebServer library, providing real-time control of appliances from any browser on the same network. For communication, the SMTP library is integrated into the ESP32 code to enable email sending directly from the microcontroller. This combination of tools ensures the system remains platform-independent, low-cost, and highly functional even in environments without internet access.

## VIII. MATHEMATICAL EQUATIONS

### 1. VOICE SIGNAL SAMPLING (NYQUIST THEOREM)

To digitize a voice signal, the sampling frequency must be at least twice the maximum frequency component:

$$f_s \geq 2 f_{\text{max}}$$

Where:

- $f_s$ = Sampling frequency
- $f_{\text{max}}$ = Maximum frequency of voice signal ($\approx 4$ kHz)

### 2. ACCURACY OF SPEECH RECOGNITION

$$\text{Accuracy} = \left( \frac{N - E}{N} \right) \times 100\%$$

Where:

- $N$ = Total number of commands
- $E$ = Number of recognition errors

### 3. POWER CONSUMPTION

$$P = V \times I$$

Where:

- $P$ = Power (watts)
- $V$ = Voltage (volts)
- $I$ = Current (amperes)

### 4. NETWORK TRANSMISSION DELAY

$$\text{Delay} = \frac{\text{Data Size}}{\text{Bandwidth}} + \text{Propagation Delay}$$

Where:

- Data Size (bits)
- Bandwidth (bits/sec)
- Propagation Delay = $\frac{\text{Distance}}{\text{Signal Speed}}$

### 5. RELAY SWITCHING TIME

$$t_{\text{total}} = t_{\text{operate}} + t_{\text{release}}$$

Typical values:

- $t_{\text{operate}} \approx$ 10–20 ms
- $t_{\text{release}} \approx$ 5–10 ms

### 6. EMAIL TRANSMISSION SUCCESS RATE

$$\text{Success Rate} = \left( \frac{N_s}{N_t} \right) \times 100\%$$

Where:

- $N_s$ = Number of successfully sent emails
- $N_t$ = Total number of send attempts

## IX. RESULTS AND DISCUSSION

The developed home automation system was tested under various practical conditions to evaluate its performance, responsiveness, and usability. The voice control functionality showed over 85% accuracy in quiet indoor environments, with an average response time of 1–2 seconds between speech input and device activation. However, accuracy slightly decreased in noisy settings, indicating the need for noise filtering enhancements. The web dashboard, hosted on the ESP32, successfully allowed real-time manual control of appliances with sub-second switching latency, and remained stable even with multiple devices connected simultaneously. The email functionality, triggered through either voice or web interface, reliably sent messages using the SMTP protocol, with delivery confirmation observed within 5 to 10 seconds. Text-to-speech

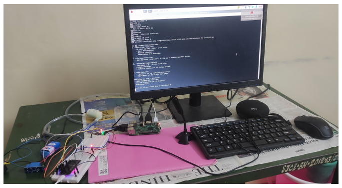

Fig. 2: Enter Caption

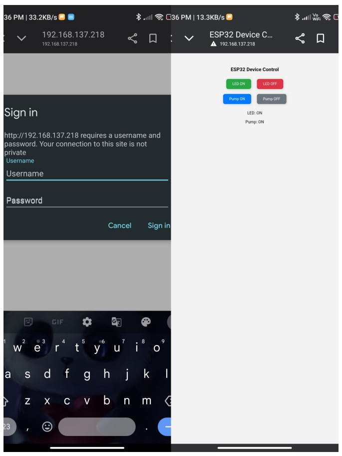


Fig. 3: Enter Caption

audio feedback enhanced interactivity, especially for elderly users. Overall, the system demonstrated reliable dual-mode operation, low power consumption, and uninterrupted performance during prolonged usage, validating its effectiveness for smart home applications.

## X. FUTURESCOPE

The current system demonstrates a functional and cost-effective smart home automation solution; however, several enhancements can be made to improve its usability, scalability, and reliability. Future developments may include the integration of multi-language voice recognition to support diverse user groups, especially in multilingual regions. A dedicated mobile application with remote access capability could be developed to allow users to control appliances beyond the local network. To enhance security, features like user authentication, web interface encryption, and secure SMTP protocols should be implemented. Additionally, the system can be extended to include sensor-based automation, such as motion detection, temperature control, or energy usage monitoring. Incorporating cloud-based data logging, AI-based command learning, and automatic software updates would further enhance system intelligence and personalization. Finally, the hardware design can be refined into a compact PCB prototype for better portability and commercial deployment.

## XI. CONCLUSION

The proposed voice-controlled home automation system successfully demonstrates the integration of offline speech recognition, web-based control, and email communication using affordable hardware such as the ESP32 and Raspberry Pi. The system achieves its primary objective of providing a dual-mode control interface—voice and browser-based—that ensures usability even in the absence of internet connectivity. Field testing confirmed reliable performance, with accurate voice command recognition, fast response times, and seamless email transmission. The inclusion of text-to-speech feedback further enhances the system's accessibility, particularly for elderly or visually impaired users. The modular architecture and use of open-source platforms make it an ideal educational project and a scalable solution for real-world smart environments. While minor limitations like reduced accuracy in noisy conditions and lack of encryption were observed, the system lays a strong foundation for future enhancements, offering a cost-effective, inclusive, and practical approach to smart home automation.

## XII. REFERENCES

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